

Measuring the physical world

Measurement connects mathematics to real objects and events. Year 3 pupils measure, compare, add and subtract length, mass and capacity, work with mixed units, and calculate the perimeter of simple 2-D shapes.

Choosing suitable units

LENGTH

mm cm m

parcel, height, distance

MASS

g kg

ingredients and parcels

CAPACITY

ml l

medicine and drinks

Useful equivalences

1 metre = 100 centimetres

1 centimetre = 10 millimetres

1 kilogram = 1,000 grams

1 litre = 1,000 millilitres

Compare and calculate with measures

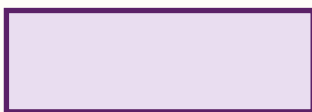
COMPARE **1 m 20 cm > 95 cm**

ADD MASS **1 kg 200 g + 300 g = 1 kg 500 g**

SUBTRACT **750 ml - 250 ml = 500 ml**

Perimeter - the distance around a shape

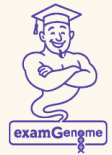
3 cm



6 cm

$$6 + 3 + 6 + 3 = 18 \text{ cm}$$

Add every outside edge once.



Measurement in everyday life

Money - totals and change

Pupils add and subtract amounts using pounds and pence, recording the units clearly. This is used when comparing prices, planning spending and checking change.

CHANGE

$$\text{£5 and 0p} - \text{£3 and 60p} = \text{£1 and 40p}$$

Reading analogue and digital time



3:20 pm

Roman numerals I-XII

Read to the nearest minute

12-HOUR AND 24-HOUR TIME

15:20 = 3:20 pm

am: before noon pm: after noon

Calculating and comparing durations

JOURNEY TIME

9:35 am → 10:20 am

25 min + 20 min = 45 minutes

TIME FACT

60 seconds = 1 minute

Use seconds, minutes and hours

Calendar facts

30 DAYS

Apr, Jun, Sep, Nov

31 DAYS

seven other months

FEBRUARY

28 days; 29 in a leap year

A YEAR

365 days; leap year 366

Pupils should be able to:

- choose and use suitable metric units
- calculate the perimeter of simple shapes
- read analogue, 12-hour and 24-hour clocks
- calculate and compare durations
- compare, add and subtract measurements
- add money and calculate change
- estimate and read time to the nearest minute
- recall key calendar facts